

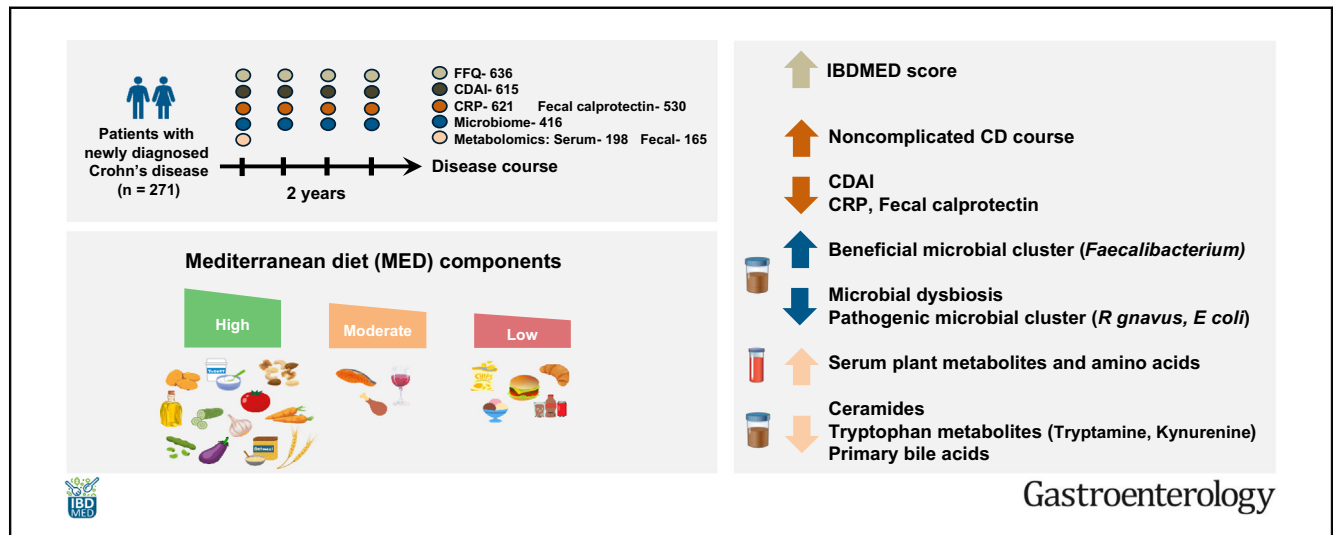
Mechanistic Implications of the Mediterranean Diet in Patients With Newly Diagnosed Crohn's Disease: Multiomic Results From a Prospective Cohort



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INFLAMMATORY
BOWEL DISEASE



See editorial on page 872.

BACKGROUND & AIMS: To decipher the mechanisms underlying the protective role of the Mediterranean diet (MED) in Crohn's disease (CD), we explored the implications of adherence to MED on CD course, inflammatory markers, and microbial and metabolite composition. **METHODS:** Patients with newly diagnosed CD were recruited and followed prospectively. MED adherence was assessed by repeated food frequency questionnaires (FFQs) using a predefined inflammatory bowel disease Mediterranean diet score (IBDMED score), alongside validated MED adherence screeners. Crohn's disease activity index (CDAl), C-reactive protein, fecal calprotectin, and microbial composition (16S-ribosomal RNA sequencing) were assessed each visit. Baseline serum and fecal samples were analyzed for targeted quantitative metabolomics. **RESULTS:** Consecutive patients: 271 (52% men, average age 31 ± 12 years, B1 phenotype 75%). FFQs

collected: 636 (range 1–5 FFQs per patient). Adherence to MED was associated with a noncomplicated CD course, and inversely correlated with CDAl, fecal calprotectin, C-reactive protein, and microbial dysbiosis index (all $P < .05$). Increasing adherence to MED over time correlated with reduced CDAl and inflammatory markers ($P < .05$). Adherence to MED correlated with a microbial cluster of commensals and short-chain fatty acid producers including *Faecalibacterium*, and with plant metabolites, vitamin derivatives, and amino acids. Conversely, adherence to MED inversely correlated with a cluster of oral genera, *Escherichia coli* and *Ruminococcus gnavus*, known CD-associated species, and with tryptophan metabolites, ceramides, and primary bile acids (false discovery rate < 0.2). **CONCLUSION:** Adherence to MED is associated with beneficial clinical outcomes and decreased inflammatory markers. These may be driven by lower levels of primary bile acids and microbial dysbiosis and a beneficial microbial and metabolite composition. Randomized controlled trials are needed to evaluate the role of MED in CD management.

Keywords: Inflammatory Bowel Disease; Nutrition; Metabolomics; Microbiome; Dysbiosis.

Inflammatory bowel diseases (IBD), including Crohn's disease (CD) and ulcerative colitis, are complex multifactorial diseases with a steadily increasing global prevalence.¹ Although its exact etiology remains elusive, mounting evidence highlights the crucial role of environmental factors in its pathogenesis.² Among these factors, diet and the gut microbiome have garnered substantial attention as modifiable risk determinants that may be targeted to affect disease course.³

Alterations in the composition, diversity, and functionality of the gut microbial community overall designated as dysbiosis are a hallmark of IBD.^{4,5} Specifically, reduced bacterial diversity, often accompanied by reduced abundances of protective gut microbial taxa such as Firmicutes, encompassing butyrate-producing bacteria like *Faecalibacterium prausnitzii*,⁶ was consistently reported. Concurrently, there is an expansion of potentially inflammatory groups, including the phylum of Proteobacteria, species of *Fusobacterium* and *Ruminococcus gnavus*.⁷⁻⁹

Microbiota composition is largely affected by diet.⁵ The Western diet, which is high in fat, sugar, animal protein, and ultra-processed foods (UPFs), has been associated with adverse modifications in microbiota composition, including reduced microbial diversity and expansion of potentially pathogenic bacterial taxa, which may contribute to inflammation.¹⁰⁻¹² As opposed to the Western diet, the Mediterranean diet (MED), which is a plant-based high-fiber diet, rich in vegetables, fruits, whole grains, nuts, legumes, and olive oil, and low in animal-based foods, specifically red meat, and UPFs, might contribute to beneficial microbial alterations.¹³ MED is associated with increased microbial diversity, and increased abundance of butyrate-producing bacteria in healthy subjects,¹⁴⁻¹⁶ patients with quiescent ulcerative colitis,¹⁷ and first-degree relatives of patients with IBD.¹⁸ MED is also associated with a variety of outcomes in patients with IBD, including inverse association with activity indices and inflammatory markers^{17,19,20} and improved metabolic status and quality of life^{21,22}; however, how MED affects disease course, inflammatory markers, and microbial composition in patients with newly diagnosed CD remains largely unclear. Here we prospectively followed a cohort of patients with newly diagnosed CD and investigated how adherence to MED, assessed by a novel microbiota-targeted MED-based adherence screener we developed, associated with disease course, inflammatory markers, and microbial and metabolite composition.

Methods

Population

This prospective observational longitudinal study was conducted in 2 tertiary IBD centers in Israel (Rabin Medical Center Beilinson Campus, Petah-Tikva, Israel and Tel Aviv Sourasky Medical Center, Tel Aviv, Israel). Consecutive patients aged ≥ 18 years old with either suspected CD or newly

WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

The Mediterranean diet has been suggested as a strategy in the prevention and management of patients with inflammatory bowel disease. The underlying mechanisms behind its potential beneficial effects are vaguely understood.

NEW FINDINGS

Adherence to the Mediterranean diet is associated with noncomplicated Crohn's disease course and lower levels of inflammatory and microbial dysbiosis markers. The Mediterranean diet and consumption of its recommended dietary components correlated with improved microbial and metabolite composition.

LIMITATIONS

This was an observational study; therefore, causality cannot be determined. It may be easier for patients with Crohn's disease and a noncomplicated course to follow a Mediterranean lifestyle. Generalizability to other populations should be assessed.

CLINICAL RESEARCH RELEVANCE

This study associates the Mediterranean diet with beneficial clinical outcomes in patients with Crohn's disease. Increasing adherence to the Mediterranean diet over time is feasible and is associated with reduction in inflammatory markers.

BASIC RESEARCH RELEVANCE

The potential beneficial effect of the Mediterranean diet can be mediated by shifting microbial and metabolite composition toward a healthier milieu, characterized by higher levels of butyrate producers and lower levels of microbial dysbiosis and pathobionts like *Ruminococcus gnavus* and inflammatory-associated metabolites.

diagnosed CD (within 6 months before enrollment) who agreed to participate were recruited and followed up over the years 2013 to 2024. CD diagnosis was confirmed based on accepted clinical, laboratory, endoscopic, histologic, and imaging criteria. Patients were managed and followed by the IBD multidisciplinary team (MDT) including IBD-oriented gastroenterologists, colorectal surgeons, IBD nurses, and IBD dietitians as previously described.^{23,24} Patient-focused weekly discussions were conducted. Patients were offered to visit the dietitian following their first study visit and after CD diagnosis was confirmed. Tailored nutritional plans including disease-specific dietary recommendations based on MED principles were provided by IBD dietitians, adapted to address disease activity, manage symptoms, and optimize nutritional status. Follow-up visits

Abbreviations used in this paper: CD, Crohn's disease; CDAI, Crohn's disease activity index; CRP, C-reactive protein; FDR, false discovery rate; FFQs, food frequency questionnaires; IBD, inflammatory bowel disease; IBDMED, inflammatory bowel disease Mediterranean diet score; IMEDAS, Israeli Mediterranean diet adherence screener; IQR, interquartile range; MDI, microbial dysbiosis index; MED, Mediterranean diet; MEDAS, Mediterranean diet adherence screener; UPF, ultra-processed food.

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were offered as needed, specifically for patients on exclusion diets, at risk for malnutrition, or requiring dietary supplementation. Although dietary approaches were used according to patient needs and preferences, the basis behind the recommendations provided was MED. Study dietitians were available between visits via emails and phone.

Clinical and Inflammatory Markers

Disease activity was determined by the Crohn's disease activity index (CDAI), on recruitment, and on each follow-up visit. Inflammatory biomarkers, specifically fecal calprotectin and C-reactive protein (CRP), were assessed on each visit as well.

Disease Course

Patients were defined as having a complicated course if they were hospitalized (excluding hospitalization at disease diagnosis), underwent IBD-related surgery, were steroid dependent (>3 months or more than 1 course in a year), required ≥ 2 biologics in a year, had a complicated perianal disease or progressed from B1 to B2/3, at baseline or throughout the follow-up. Patients with at least 1 year of follow-up without any of the above complications were classified as having a noncomplicated course.

Dietary Assessment

Dietary consumption of food groups and nutrients was evaluated using food frequency questionnaires (FFQs) adapted to the Israeli population and administrated in "MABAT" Israeli nutrition and public health governmental study.²⁴ FFQs were collected at study visits that were separated from the dietitian visits. This was done to minimize the risk for social desirability bias, in which patients will answer as they think it is expected from them, following the recommendations they received. For each of the food items, a commonly used portion size was specified. Patients were asked how often on average, they consumed each food item, their dietary habits, food avoidances, use of nutritional supplements, as well as physical activity and smoking habits. Daily intake of nutrients was calculated by multiplying the frequency of each food item by the nutrient content based on the US Department of Agriculture food database and the Israeli food database "TZAMERET." Questionnaires with >5000 or <500 kcal/d were excluded. In addition, we excluded FFQs derived from patients on exclusive enteral nutrition.

MED Adherence Scores

To better assess diet-microbiome interactions in patients with IBD, we developed a novel microbiota-targeted MED-based adherence screener (IBDMED) based on the previously validated MED adherence screeners: Mediterranean diet adherence screener (MEDAS), developed in the PREDIMED study on the effect of MED on cardiovascular diseases,²⁵ and the Israeli MEDAS (IMEDAS), a culturally adapted screener that was validated in the Israeli population.²⁶ The IBDMED screener positively scored high consumption of MED-recommended dietary components such as fruits, vegetables, olive oil, legumes, nuts and seeds, and fish. It also positively scored low consumption of MED non-recommended dietary components such

as red and processed meat, soft drinks, and sweets. To this we added several dietary features based on previous data associated with microbial composition and function. These included a positive score for plant diversity²⁷ by scoring for different colors in the diet, consumption of fermented foods²⁸ (specifically yogurt), and inclusion of starchy vegetables like potato in the whole grain category to promote diversity in the carbohydrate-rich food group, as well as support butyrate producers as we had previously shown.²⁹ In addition, we aimed to positively score for relatively low UPF intake. To this end, we evaluated the average intake of sweets, snacks, sweet and savory pastries, soft drinks, and foods and drinks containing artificial sweeteners. The scoring system of the IBDMED is detailed in the [Supplementary Methods](#).

Microbial Analysis

Fecal specimens were aliquoted at the medical institute and stored at -80°C , DNA extraction and deep amplicon sequencing of bacterial 16S ribosomal RNA gene (V4 region) were performed at HyLabs (Rehovot, Israel), Next Generation Sequencing unit; raw sequencing data were processed at Gophna lab (TAU) using R package dada2 v1.22.0. Details on DNA extraction, PCR amplification, sequencing, and bioinformatics analysis are described in the [Supplementary Methods](#). Shannon diversity was assessed using the "vegan" R package v2.6-4. The microbial dysbiosis index (MDI) was calculated as described³⁰ (the ratio between abundance of bacteria that are increased in CD divided by abundance of bacteria that are decreased in CD), with a single modification of adding the IBD-associated bacterium *R. gnavus* to the numerator rather than the denominator. Fecal samples of patients who were exposed to antibiotics 3 months before sample collection were excluded from the analysis.

Metabolomic Analysis

Baseline serum and fecal samples were analyzed for metabolite composition in The Metabolomics Innovation Centre, Canada, using a targeted quantitative metabolomics approach with a combination of direct injection mass spectrometry with a reverse-phase liquid chromatography-tandem mass spectrometry custom assay. This custom assay, in combination with an ABSciex 5500/4000 QTrap (Applied Biosystems/MDS Sciex) mass spectrometer, was used for the targeted identification and quantification of up to 650 different endogenous metabolites including amino acids, acylcarnitines, biogenic amines and derivatives, uremic toxins, glycerophospholipids, sphingolipids, sugars, and fecal bile acids as detailed in the [Supplementary Methods](#).

Statistical Analysis

Statistical analysis and data visualization were conducted using R and Python and its libraries, including numpy, pandas, matplotlib, SciPy, and seaborn. The distribution of MED adherence scores among patients with complicated and noncomplicated CD courses was visualized using the hist function from the matplotlib library. Generalized estimating equation models were applied to control for repeated measurements of FFQs. Spearman's correlations and statistical analysis were conducted using the functions spearmanr from the SciPy library. To analyze changes in IBDMED scores across

time points, we used a combination of data visualization techniques including a line plot, using the matplotlib library that includes lines for individual patients, as well as median lines for both complicated and noncomplicated cases, calculated at each time point, allowing for comparison of the central trends between groups. Spearman's correlation analysis was conducted to assess the interaction between dietary components and microbial taxa. To reduce spurious correlations, microbial features (prevalence >20 and minimal relative abundance >0.01) were filtered before correlating. Spearman correlations were calculated between dietary factors and 3 types of metabolites including serum metabolites, fecal metabolites, and fecal bile acids. The Benjamini-Hochberg method was used to adjust *P* values and control for the false discovery rate (FDR). Significant correlations were identified using an FDR threshold of 0.2 and visualized in a clustered heatmap.

Ethical Consideration

The protocol was approved by local ethics committees: Rabin Medical Center Institutional Review Board 0298-17 and Tel Aviv Sourasky Medical Center Institutional Review Board (0467-10). Written informed consent was obtained from all participants.

Results

Demographics and Disease Course

We recruited 271 consecutive patients with newly diagnosed CD. Median age was 27 years (interquartile range [IQR] 22–38), 48% were women, and 43% were Ashkenazi Jews. Median body mass index was 21.9 (IQR 20.0–25.3), and 17% were active smokers. Most patients presented with ileal or ileocolonic disease, and the inflammatory phenotype (B1) was indicated in 76% of patients. Demographic and clinical characteristics are presented in Table 1.

A total of 215 patients were evaluated every 6 months and followed up for a median time of 24 months (IQR 12–37). Throughout the follow-up, 146 (68%) patients received biologic therapy with anti-tumor necrosis factor (anti-TNF) as the most common agent, 111 (52%) received steroids, mostly budesonide, 87 (40%) immunomodulators, and 68 (32%) 5-aminosalicylic acid. The main reasons for dropout were distance from the hospital, the COVID-19 pandemic, and the burden of study requirements (samples and questionnaires).

Overall, 105 (49%) patients had a noncomplicated course throughout the follow-up period, and 110 (51%) had a complicated course that was defined by IBD-related hospitalization or surgery, steroid dependency, ≥2 biologic drugs in 1 year, phenotype progression, or development of complex perianal disease. Of note, in the vast majority (105 of 110 [95%]) of patients with a complicated course, complication occurred within the first year.

Dietary Habits

Dietary habits of patients were assessed by FFQs. We collected a total of 636 from 271 patients (range 1–5 FFQ per patient, Supplementary Table 1). The average daily

Table 1. Demographic and Clinical Characteristics of the Study Cohort

N = 271	
Sex, female, n (%)	131 (48)
Age, median (IQR)	27 (22–38)
BMI (kg/m ²), median (IQR)	21.9 (20–25.3)
Active smokers, n (%)	46 (17)
Family history of IBD, n (%)	76 (28)
Origin, Ashkenazi, n (%)	116 (43)
Disease location, ^a n (%)	
Ileal (L1)	141 (53)
Colonic (L2)	39 (15)
Ileocolonic (L3)	87 (32)
Upper GI (L4)	35 (13)
Perianal (P)	58 (21)
Disease phenotype ^b	
Inflammatory (B1), n (%)	202 (76)
Stricturing (B2), n (%)	40 (15)
Penetrating (B3), n (%)	23 (9)
CDAI, median (IQR)	133 (65–230)
CRP (mg/dL), median (IQR)	0.9 (0.3–2.2)
Fecal calprotectin (μg/g), median (IQR)	420 (156–1300)
Disease course (N = 215)	
Follow-up time interval (mo), median (IQR)	24 (12–37)
Medications, n (%)	
Steroids	111 (52)
5-ASA	68 (32)
IMM	87 (40)
Biologics	146 (68)
Disease complications, n (%)	
CD-related hospitalization	57 (26)
CD-related surgery	32 (15)
Steroid dependency	39 (18)
≥2 biologics	14 (6.5)
Complex perianal disease	35 (16)
Change in disease phenotype	9 (4)
Noncomplicated in the first year	105 (49)

5-ASA, 5-aminosalicylic acid; BMI, body mass index; IMM, immunomodulators.

^aData available for 267 patients.

^bData available for 265 patients, assessed at study enrollment.

caloric intake of patients was 2018 kcal/d, with 21% of kcal derived from protein, 34% from fat, and 45% from carbohydrates (Supplementary Table 2). The average daily fiber intake was 21 g/d. Notably this is above the average fiber intake of 16.2 g/d in the American population, but below the general recommendation for the healthy population (25–38 g/d).^{31,32}

To assess diet-microbiome interactions in patients with IBD, we developed the IBDMED score. Here, in addition to assessment of MED-recommended and non-recommended dietary components (see Methods), we also assessed for dietary factors that were previously linked to microbial composition like plant diversity, assessed by diversity of colors from plant sources in the diet and fermented food consumption, that were previously linked to increased microbial diversity.²⁸ For each FFQ, we calculated adherence

to MED scores using the MEDAS, IMEDAS, and IBDMED. The 3 scores highly correlated with each other ($r > 0.69$, $P < .05$, [Supplementary Figure 1](#)). The median MEDAS, IMEDAS, and IBDMED scores were 6 (5–7), 8 (6–9), and 8 (7–11), respectively. This was comparable overall to previous reported data for the healthy Israeli population.²⁶ Patients consumed 3 servings of vegetables and 1 serving of fruit per day, with red (mainly tomato) and green (mainly cucumber) as main colors consumed daily. Overall intake of whole grains, legumes, nuts and seeds, and olive oil was relatively low. Sweets were the most common non-recommended dietary components with median intake of 0.8 servings/day followed by red and processed meat with 0.5 servings/day ([Supplementary Table 3](#)).

MED Was Inversely Associated With Clinical, Inflammatory, and Dysbiosis Markers

We first asked whether adherence to MED is associated with disease course, clinical, inflammatory, and microbial markers. The average adherence to MED over time was higher in patients with noncomplicated vs complicated course, as presented by the 3 scores ($P < .05$ for all, [Figure 1A](#)); this remained significant when controlling for repeated measurements ($P < .05$, [Supplementary Table 4](#)) or when taking into account only baseline data ([Supplementary Figure 2](#)). In the entire cohort, adherence to MED was inversely associated with CDAI, CRP, fecal

calprotectin, and microbial dysbiosis ($P < .05$ for all). No association was demonstrated with Shannon α -diversity. Following patients' stratification according to disease course, patients with complicated course presented an inverse association between adherence to MED and inflammatory markers, whereas patients with noncomplicated course presented an inverse association between adherence to MED and microbial dysbiosis ($P < .05$, [Figure 1B](#), [Supplementary Table 5](#)). Notably, when stratifying patients according to disease behavior and location, the inverse associations were noted more in patients with the inflammatory phenotype, ileal disease, and no perianal involvement. However, in patients with fibrostenotic disease, we detected inverse associations with inflammatory and dysbiosis markers and in patients with colonic disease, inverse associations with clinical and inflammatory markers. Medical treatments also mediated these interactions, as reflected by adherence to MED inverse correlation with clinical and inflammatory markers in patients on biologics and immunomodulators and with dysbiosis in patients with no advanced treatments ($P < .05$, [Figure 2](#)).

We then asked whether patients with newly diagnosed CD are changing their diet over time and whether changes in the diet are associated with changes in clinical, inflammatory, and microbial markers. To this end, we analyzed repeated measurements of dietary, clinical, inflammatory, and microbial assessments (time interval between visits 6.8 months [IQR 6.0–12.3]). We noted inter- and intra-

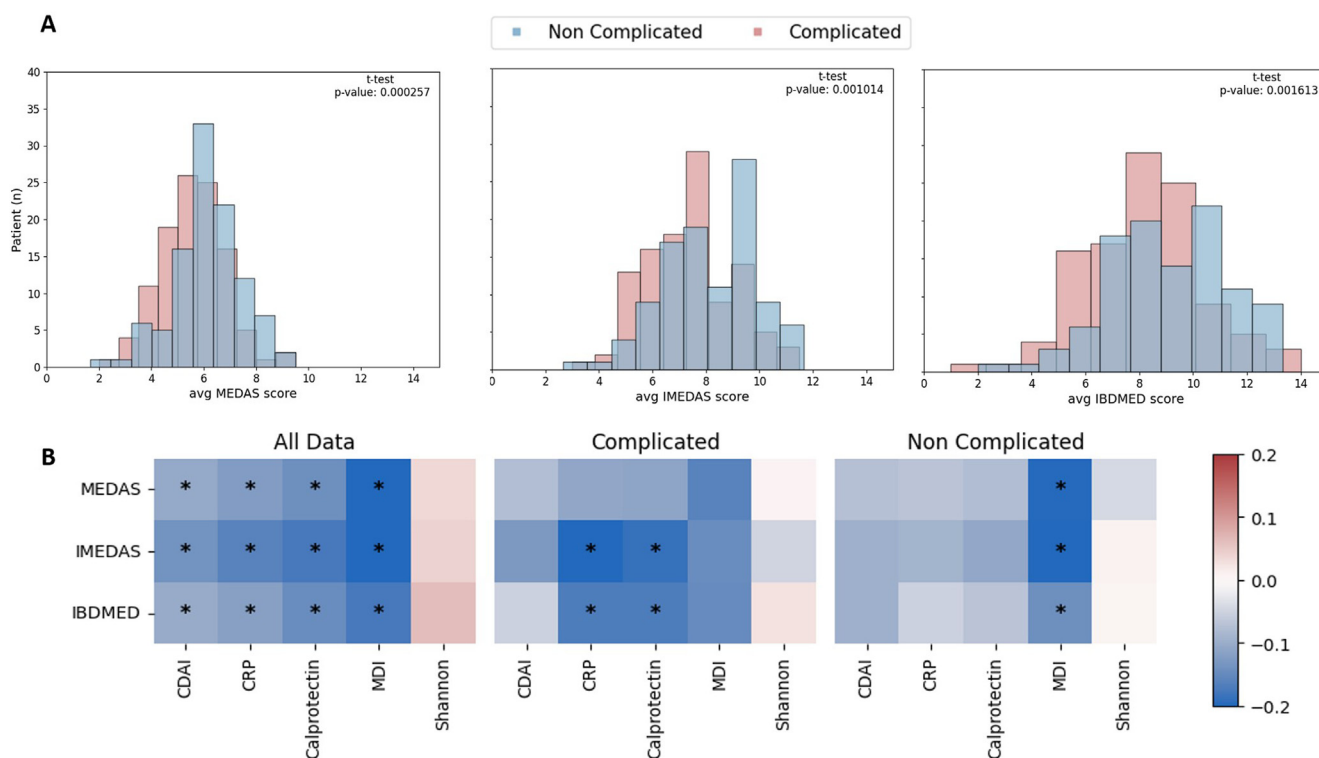


Figure 1. Adherence to MED is inversely associated with clinical, inflammatory and microbial dysbiosis markers in patients with newly diagnosed CD. (A) Distribution of average MED adherence scores among patients with complicated (n = 110) vs noncomplicated (n = 105) CD course. (B) Spearman's correlations between MED adherence scores and clinical (CDAI), inflammatory (CRP, calprotectin), and microbial (MDI, Shannon diversity) scores across patients with complicated vs noncomplicated CD course. * $P < .05$ (all data: n = 271).

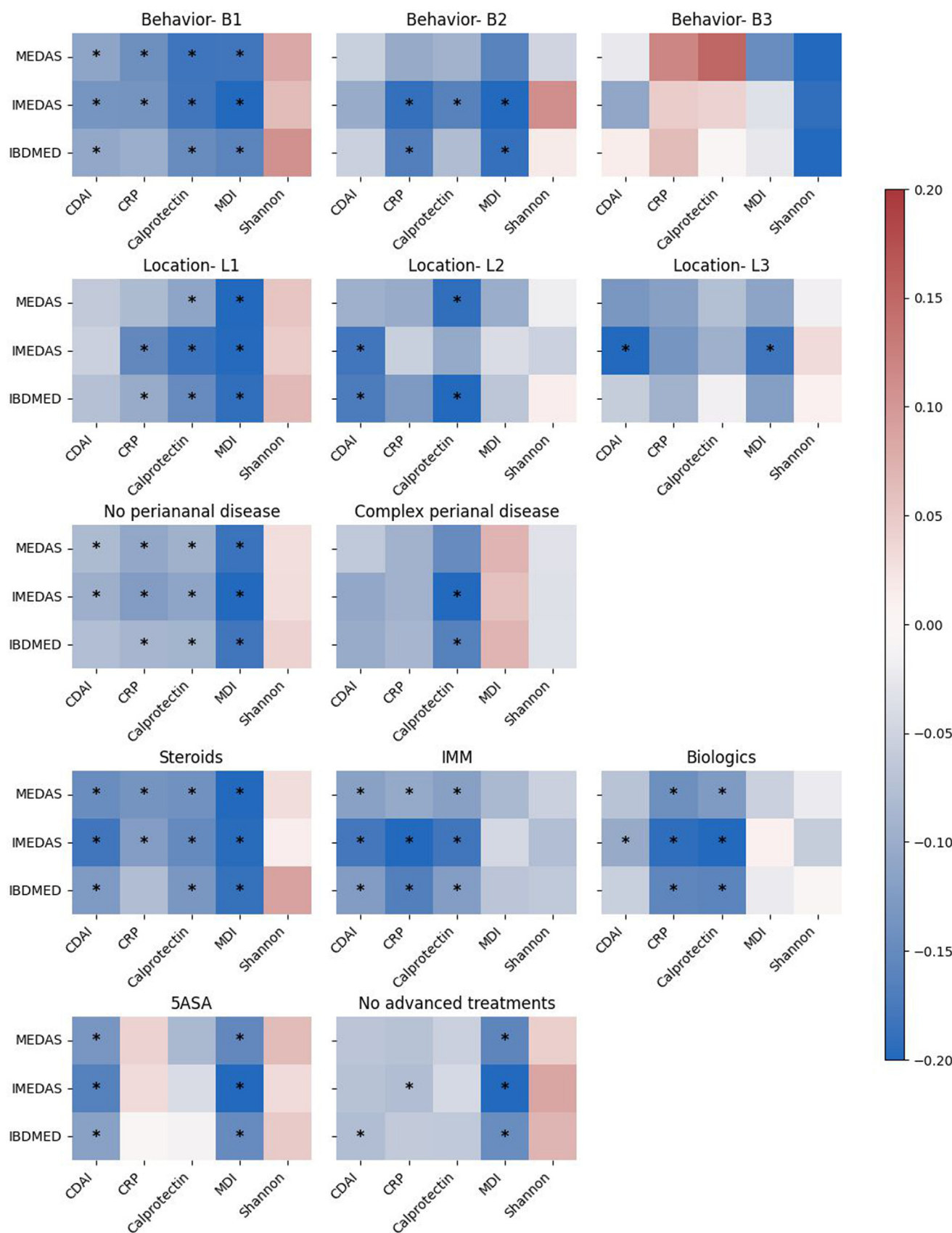


Figure 2. Disease phenotype and medical therapy mediate the interactions between adherence to MED and clinical, inflammatory, and dysbiosis markers. Spearman's correlations between MED adherence scores and clinical (CDAl), inflammatory (CRP, calprotectin), and microbial (MDI, Shannon diversity) scores stratified according to disease behavior (B1: inflammatory, B2: fibrosenotic, B3: penetrating); location (L1: terminal ileum, L2: colon, L3: ileocolonic), and medical treatments. * $P < .05$. 5-ASA, 5-aminosalicylic acid; IMM, immunomodulators.

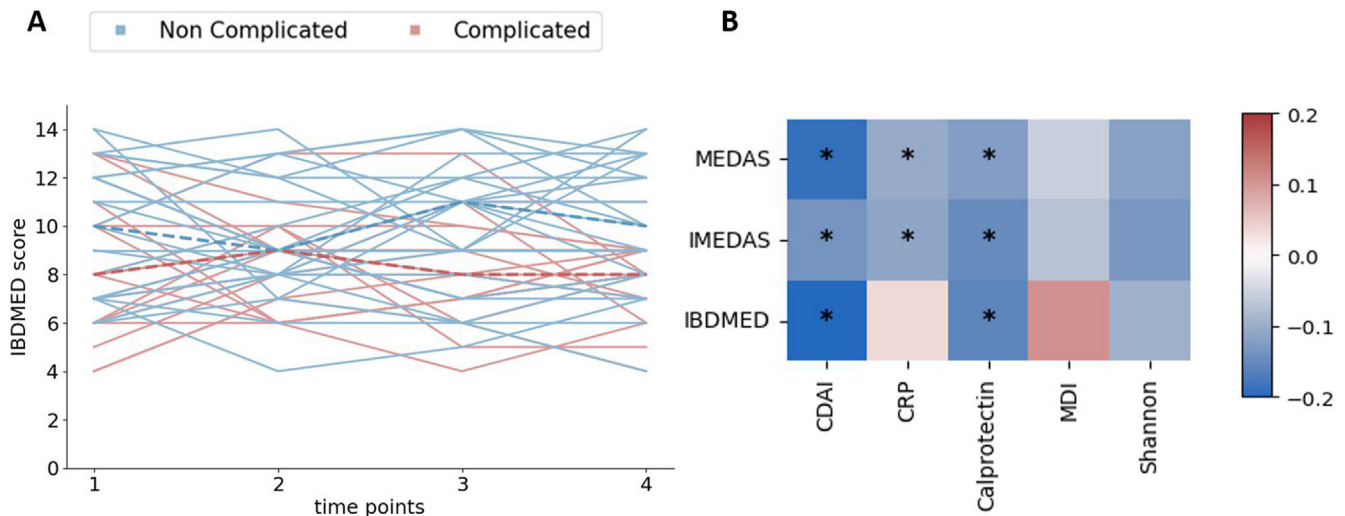


Figure 3. Increasing adherence to MED over time is associated with decreased clinical and inflammatory markers. (A) Changes in IBDMED score across 4 time points in 24 patients with noncomplicated and 20 patients with complicated CD course. (B) Spearman's correlations between changes (deltas) in MED adherence scores and changes (deltas) in clinical (CDAI), inflammatory (CRP, calprotectin), and microbial (MDI, Shannon diversity) markers. * $P < .05$.

individual variability in dietary intake assessed by the IBDMED score in 24 patients with noncomplicated course and 20 with complicated course across 4 consecutive time points. Although overall adherence to MED was comparable across time points (Figure 3A, Supplementary Table 6), increasing adherence to MED over time was associated with decreasing clinical and inflammatory markers (Figure 3B, Supplementary Table 7). In a subset of patients with FFQs collected before and after dietary counseling, variable changes in adherence to MED were observed, with deltas ranging from 7 points decrease and 8 points increase (Supplementary Figure 3A). The overall adherence to MED before and after dietary counseling increased from 8.0 ± 2.8 to 9.0 ± 3.1 ($P = .008$, Supplementary Figure 3B).

Next, we assessed the contribution of specific dietary components to the interactions between diet and inflammatory and microbial markers. We identified an overall pattern demonstrating that clinical, inflammatory, and microbial dysbiosis markers were inversely associated with MED-recommended dietary components and positively associated with MED non-recommended dietary components (Figure 4, Supplementary Table 8). The most responsive marker was the MDI, which showed inverse associations with vegetables, fruits, legumes, nuts and seeds, different plant food colors, fish, and red wine. Clinical and inflammatory markers were inversely associated with vegetables, whole grains and starchy vegetables, legumes, nuts and seeds, red wine, and specifically with red, yellow, and purple colors and positively with soft drinks, snacks, and pastries. Of note, patients with noncomplicated course presented more significant correlations between dietary components and clinical, inflammatory, and microbial dysbiosis markers compared with patients with complicated course. This observation was even stronger when only baseline data were assessed (Supplementary Figure 4).

MED Correlated Positively With Beneficial and Inversely With Pathogenic Microbial Clusters

We then focused on the interactions between dietary components and microbial genera. We identified 2 dietary and 2 microbial clusters: one dietary cluster was composed of MED-recommended dietary components and the IBDMED score and the other was composed of MED non-recommended dietary components. One microbial cluster was composed of commensals typical to the non-IBD microbiome and short-chain fatty acid producers including *Faecalibacterium*, thus designated from here on "beneficial"; the other was enriched in oral genera and also included *Escherichia coli* and *R. gnavus*, previously associated with CD, thus designated "pathogenic." The dietary cluster of MED-recommended dietary components displayed more correlations with microbial genera compared with the dietary cluster of the non-recommended dietary components, suggesting that the foods that are included in the diet might have a greater effect on the microbiome compared with the foods that are recommended to be excluded from it. The cluster of MED-recommended dietary components positively correlated with the beneficial microbial cluster, and inversely with the pathogenic. Specifically, fruits, nuts and seeds (derived mainly from tahini), and yellow color were the food groups that mostly correlated with the beneficial microbial cluster, whereas vegetables and green color displayed more inverse correlations with the pathogenic microbial cluster. The IBDMED score positively correlated with 5 genera from the beneficial microbial cluster and inversely with *E. coli* and *R. gnavus* (Figure 5, Supplementary Table 9).

MED Interacted With Inflammatory-Associated Fecal and Serum Metabolites

Finally, we assessed the interactions between diet and fecal and serum metabolites. Baseline serum and fecal

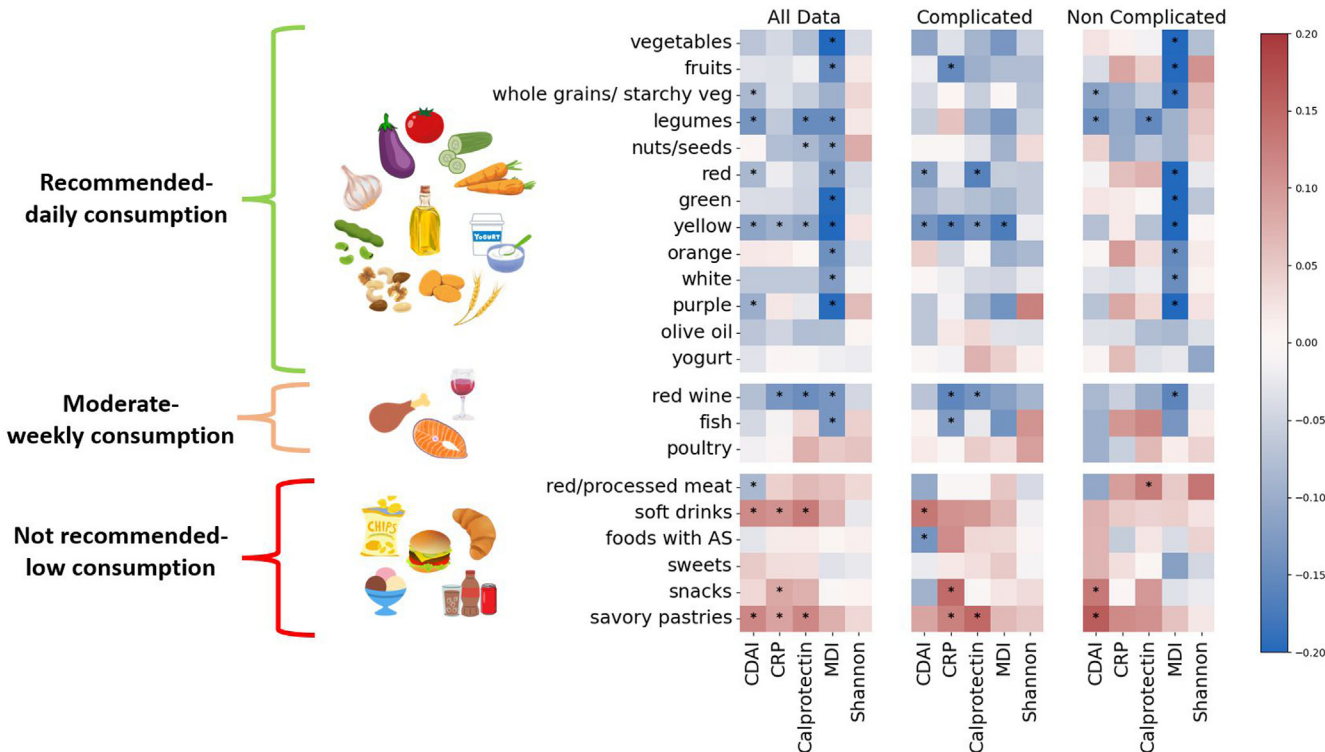


Figure 4. MED-recommended dietary components are inversely associated with clinical, inflammatory, and dysbiosis markers in patients with newly diagnosed CD. Spearman's correlations between MED-recommended (green recommended for daily consumption and orange for weekly consumption) and non-recommended (red for limited/ no consumption) dietary components and clinical (CDAI), inflammatory (CRP, calprotectin), and microbial (MDI, Shannon diversity) scores across patients with complicated vs noncomplicated CD course. * $P < .05$. AS, artificial sweeteners.

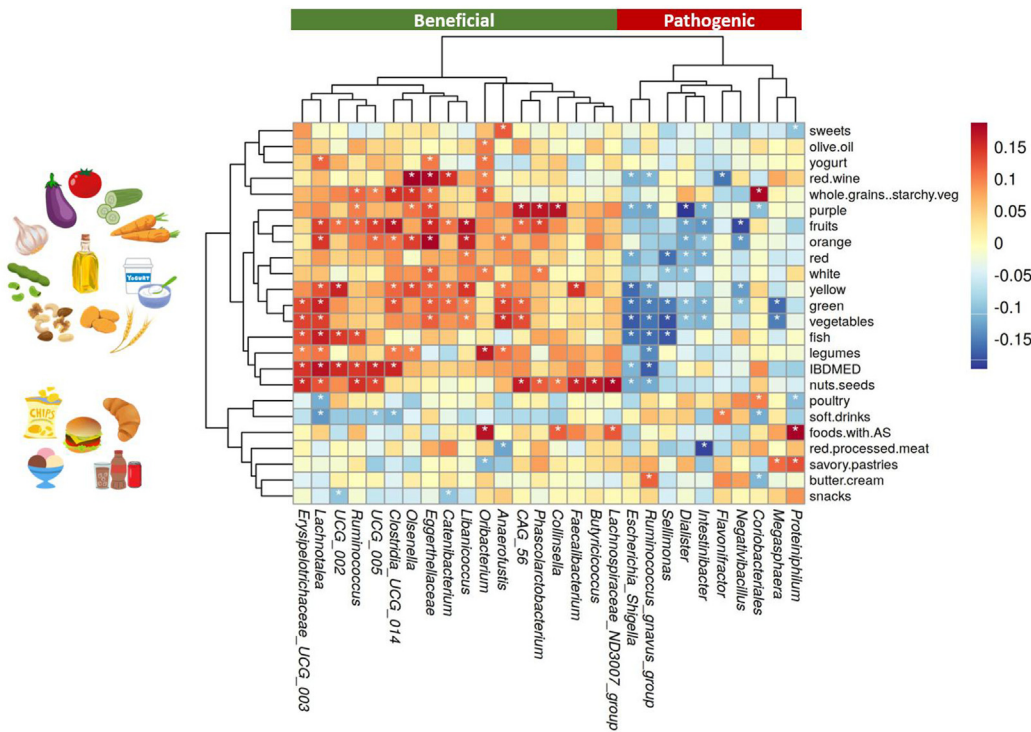


Figure 5. Adherence to MED and its recommended dietary components is positively associated with beneficial, and inversely with pathogenic microbial clusters. Spearman's correlations between MED-recommended and non-recommended dietary components and microbial taxa in patients with available microbial analysis ($n = 219$ patients with 406 samples). Patients who were exposed to antibiotics 1 month before sample collection were excluded from the analysis. Only taxa that had at least 1 correlation with $q < 0.2$, $P < .05$, and $|\text{Rho}| > 0.15$ are shown. AS, artificial sweeteners.

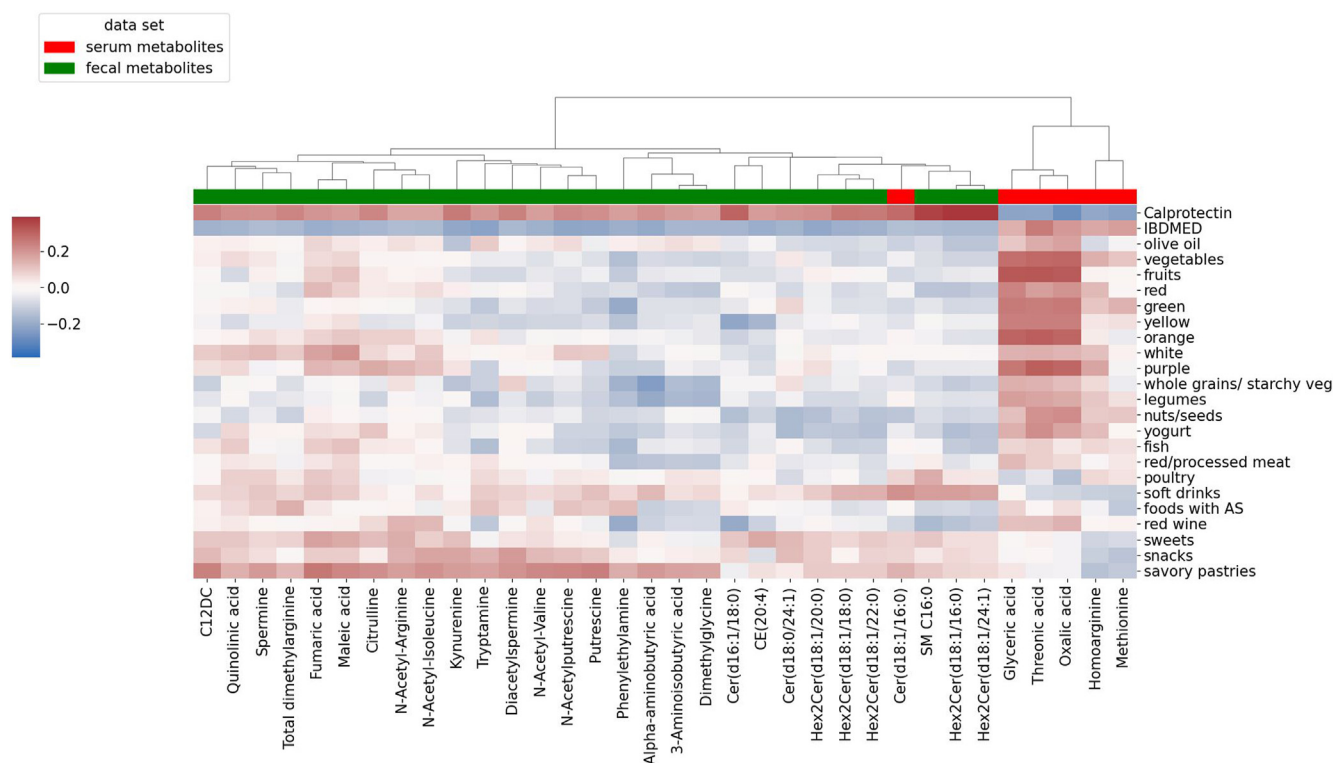


Figure 6. Discordant associations between MED-recommended vs non-recommended dietary components and metabolite clusters. Spearman's correlations between MED-recommended and non-recommended dietary components and serum metabolites (red, $n = 198$ patients), fecal metabolites (green, $n = 165$ patients) assessed at baseline. Only metabolites that were significantly associated with fecal calprotectin ($FDR < 0.2$) and with the IBDMED score are presented. AS, artificial sweeteners; CE, cholesterol esters; Cer, Ceramides.

samples were assessed for metabolite composition using a targeted approach. To explore the interaction between diet and metabolites that may mediate inflammation, we focused on serum and fecal metabolites that were significantly correlated with fecal calprotectin. Of 117 fecal metabolites and 55 serum metabolites that significantly correlated with fecal calprotectin ($FDR < 0.2$), 28 fecal and 6 serum metabolites significantly correlated with the IBDMED score, with all correlations detected in the opposite directions, that is, metabolites that positively correlated with fecal calprotectin, negatively correlated with the IBDMED score and vice versa (Supplementary Table 10). In a cluster analysis, correlating between serum and fecal metabolites and the IBDMED recommended and non-recommended dietary components, we identified 3 types of metabolite clusters. The first positively correlated with the IBDMED score and its recommended dietary components and negatively with fecal calprotectin. This cluster comprised serum organic compounds derived mainly from dietary plant components like glyceric and oxalic acids, vitamin C derivatives like threonic acid, and also included amino acids like methionine and homoarginine. The second cluster inversely associated with the IBDMED score and positively with fecal calprotectin and with the non-recommended dietary components, mainly sweets, snacks, and savory pastries. This cluster included fecal compounds

that are commonly used in the food industry, such as malic acid that is often used to provide a fruity sour flavor in soft drinks, jams, and candies and fumaric acid that is used as an acidity regulator in baking products. The third cluster inversely correlated with the IBDMED score and its recommended dietary components and positively with fecal calprotectin. It comprised serum and fecal ceramide metabolites, fecal cholesterol esters, tryptamine, and kynurenine, which are involved in tryptophan metabolism known to be elevated in patients with IBD³³ (Figure 6, Supplementary Table 10). Tryptamine can be produced by *R. gnavus*, which was also inversely associated with adherence to MED, thus this observation may reflect the complex interactions between dietary components, microbes and metabolites. Of note, cholic acid, the major primary bile acid in humans, as well as chenodeoxycholic acid and ursodeoxycholic acid, were also inversely correlated with the IBDMED score (Supplementary Table 10, $FDR < 0.2$).

Taken together, MED pattern and its recommended dietary components are inversely associated with complicated course, clinical, inflammatory, and microbial dysbiosis markers in patients with newly diagnosed CD. These associations might derive from interactions between the dietary components consumed and their potential effects on microbial and metabolite composition.

Discussion

In this prospective observational longitudinal follow-up study in patients with newly diagnosed CD treated by an MDT, we showed that adherence to MED was associated with a noncomplicated CD course and with lower levels of inflammation and microbial dysbiosis. Furthermore, increasing adherence to MED over time correlated with reduction in inflammatory markers. This might be mediated by MED-positive associations with beneficial microbial groups and anti-inflammatory metabolites and inverse associations with microbial dysbiosis, and inflammatory-associated microbial groups and metabolites.

MED had been recognized as a beneficial dietary approach in multiple chronic morbidities.^{34–36} In recent years, there is an increasing support for its relevance in IBD.^{37,38} MED was associated with lower rates of later-onset CD³⁹ and improved microbial composition in first-degree relatives of patients with CD,¹⁸ modest effect on CD activity,¹⁹ inverse associations with inflammatory markers,^{17,19,20} IBD complications,⁴⁰ and improvement in metabolic status and quality of life.^{21,22} Here we add that adherence to MED is associated with a noncomplicated CD course as well as the inverse associations with inflammatory and microbial dysbiosis markers across various CD phenotypes and treatments, suggesting that MED might have a role in both patients with complicated and noncomplicated CD course. Our results are in line with the recent recommendation to consider MED as an optimal nutritional approach in patients with IBD.³⁷

Adherence to MED in our cohort was comparable to the general non-IBD population in Israel,²⁶ which might be partially related to the tailored dietary counseling patients received that is overall based on MED principles, and led to increasing adherence to MED over time. To address MED dietary factors that modify the microbiome, we developed the IBDMED dietary score and calculated it based on FFQs. In addition to the positive scoring of MED-recommended dietary components, we positively scored the inclusion of fermented foods and dietary plant diversity by scoring for different colors in the diet. We also translated previous data from our group, demonstrating that starchy vegetables like potatoes increase the levels of butyrate producers in patients with IBD,²⁹ and added them to the whole grains category to increase diversity in the carbohydrate-rich category. In addition, as assessment for food processing is often lacking in adherence to MED scores, we partially considered food processing in our analysis by positively scoring for low intake of sweets, snacks, pastries, soft drinks, and foods containing artificial sweeteners. This addition differentiates the IBDMED score from previously used scores such as MEDAS and IMEDAS. Despite the overall comparable performance of the 3 scores in MED adherence assessment demonstrated hereby, we find the specific additional components of the IBDMED score such as multiple colors in the diet, which correlated with beneficial microbial groups, relevant for multiple clinical purposes including nutritional patient education and advice. The large focus on the inclusion of foods in the diet, rather than

exclusion alone, is an important approach in the era of exclusion diets in IBD and their potential consequences on patients' food-related quality of life.^{41,42}

Mechanistically, as MED is a plant-based, high-fiber diet, part of its anti-inflammatory effects might be mediated by the gut microbiome, as was previously shown in non-IBD populations.¹⁴ Here, high adherence to MED and its recommended dietary components clustered with beneficial microbial groups like *Faecalibacterium* and other butyrate-producing bacteria.^{5,43,44} It also positively correlated with specific beneficial metabolites that were inversely correlated with fecal calprotectin, specifically metabolites derived mainly from plants with known anti-inflammatory properties like glyceric acid or the vitamin C derivative threonic acid.^{45–47} Conversely, low adherence to MED and MED non-recommended dietary components clustered with microbial groups and metabolites associated with inflammation, specifically *E coli* and *R gnavus* as well as their metabolites such as tryptamine, previously shown to have a pro-inflammatory role in IBD.^{8,48,49} It also correlated with primary bile acids, tryptophan metabolites such as kynurenine and ceramides, elevated during active IBD,^{33,50} and positively correlated with fecal calprotectin. Our results are supported by previous reports focusing on the general population, specifically the positive association between MED and butyrate-producing bacteria, and the negative link with pathobionts such as *R gnavus* and inflammation-related metabolites such as primary bile acids, kynurenine, and ceramides.^{51–53}

As long-term remission rates are limited in patients with IBD, even on advanced therapies,⁵⁴ there is a need for additional strategies, such as add-on therapies with broader targets, rather than a specific cytokine or signaling pathway.⁵⁵ Such targets may include microbial dysbiosis, believed to drive intestinal inflammation. Thus, further research should focus on microbial-targeted interventions reducing dysbiosis such as MED. This may be relevant in multiple clinical scenarios, including the long-term effects of beneficial diets on the development and course of IBD in patients and their relatives¹⁸ and the potential contribution of the Western diet to CD development and course.^{56–58} Finally, dietary modification may be more acceptable and less expensive than some of the current treatments for which it may be used as an add-on or partially addressing unmet needs such as optimization, safety, and quality of life. The potential beneficial role of MED, as suggested by our results, should be assessed in randomized controlled studies to further investigate its role in CD prevention and management.

This study has several strengths: this is a large and prospective study that assessed dietary, clinical, inflammatory, and microbial parameters repeatedly over approximately 2 years of follow-up in patients with newly diagnosed CD; patients were highly phenotyped allowing us to stratify according to disease course trajectory; follow-up and management were by an MDT of IBD experts, allowing a systematic coordinated and patient-centered approach; and the focus on patients with newly diagnosed CD allowed a more homogeneous group, potentially addressing a more modifiable intestinal milieu. A novel microbiome-targeted

dietary adherence screener, IBDMED score, was developed. Using this score, we identified interactions between diet composition and diversity and microbial dysbiosis. We demonstrated the comparable performance of the IBDMED score with previously validated scores, while adding more clinically translatable and culturally generalizable dietary aspects; the integration of a comprehensive microbiome, fecal, and serum metabolite analysis allowed us to identify specific interactions that deepen our mechanistic understanding of the complex MED-host-microbiome-metabolome cross talk and the potential effect on inflammation.

A few limitations should be acknowledged: this is an observational, not interventional study, therefore reverse causality cannot be ruled out. We partially addressed this by assessing diet, clinical, inflammatory, and microbial parameters repeatedly and longitudinally allowing us to identify dynamics in these markers over time and across various disease phenotypes and treatments. To demonstrate the causality, applicability, and generalizability of MED in improving CD course, further randomized controlled studies, comparing MED with sham diet or with other local diets in patients from diverse geographical regions are needed. Notably, in a previous interventional study in healthy subjects conducted by our group,¹⁶ we demonstrated the capability of MED to improve inflammatory markers and microbial composition; FFQs, while abundantly used in nutrition studies, are prone to recall bias and are less accurate compared with methods like 7-day food diaries. Yet, FFQs may better capture long-term dietary patterns like MED, which assess weekly or monthly food group consumption and are less affected by short-term variations such as seasonality or holidays; there are multiple factors, specifically CD treatments, that can affect clinical, inflammatory, and microbial dysbiosis markers. We addressed this by stratifying patients according to CD course, phenotypes, and treatments, and were able to demonstrate the inverse association between MED and microbial dysbiosis even in a subset of patients without any advanced treatments.

In conclusion, in patients with newly diagnosed CD, we showed that adherence to MED and a noncomplicated CD course were associated, as were improved markers of inflammation, microbial dysbiosis, and metabolite composition. Microbial-targeted dietary interventions like MED warrant further assessment in randomized controlled studies as strategies to improve the intestinal microenvironment and CD course.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at <https://doi.org/10.1053/j.gastro.2024.12.031>.

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Conflicts of interest

These authors disclose the following: Lihi Godny: Takeda, Ferring, Altman. Tamar Pfeffer-Gik: Janssen, Danone, Altman. Jacob E. Ollech: Pfizer. Henit Yanai: Pfizer, Abbvie, Janssen, Takeda, Bristol Myers Squibb, Eli Lilly. Iris Dotan: Abbvie, Arena, Gilead/Galapagos, Genentech/Roche, Athos, Janssen, Pfizer, Takeda, Altman Research, Celgene/BMS, Celltrion, Eli Lilly, Dr. Falk Pharma, Ferring, Sandoz, Harp Diagnostics. The remaining authors disclose no conflicts.

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Data Availability

The data cannot be shared publicly for the privacy of individuals who participated in the study. The data will be shared on reasonable request to the corresponding author with the permission of our local institutional review board.

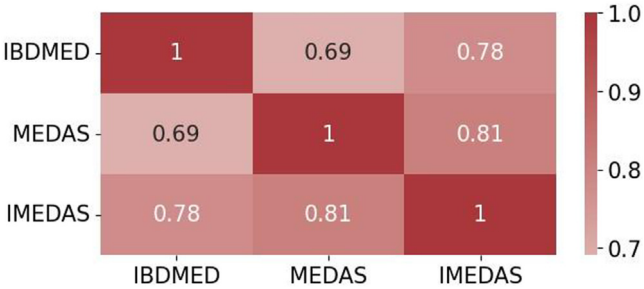
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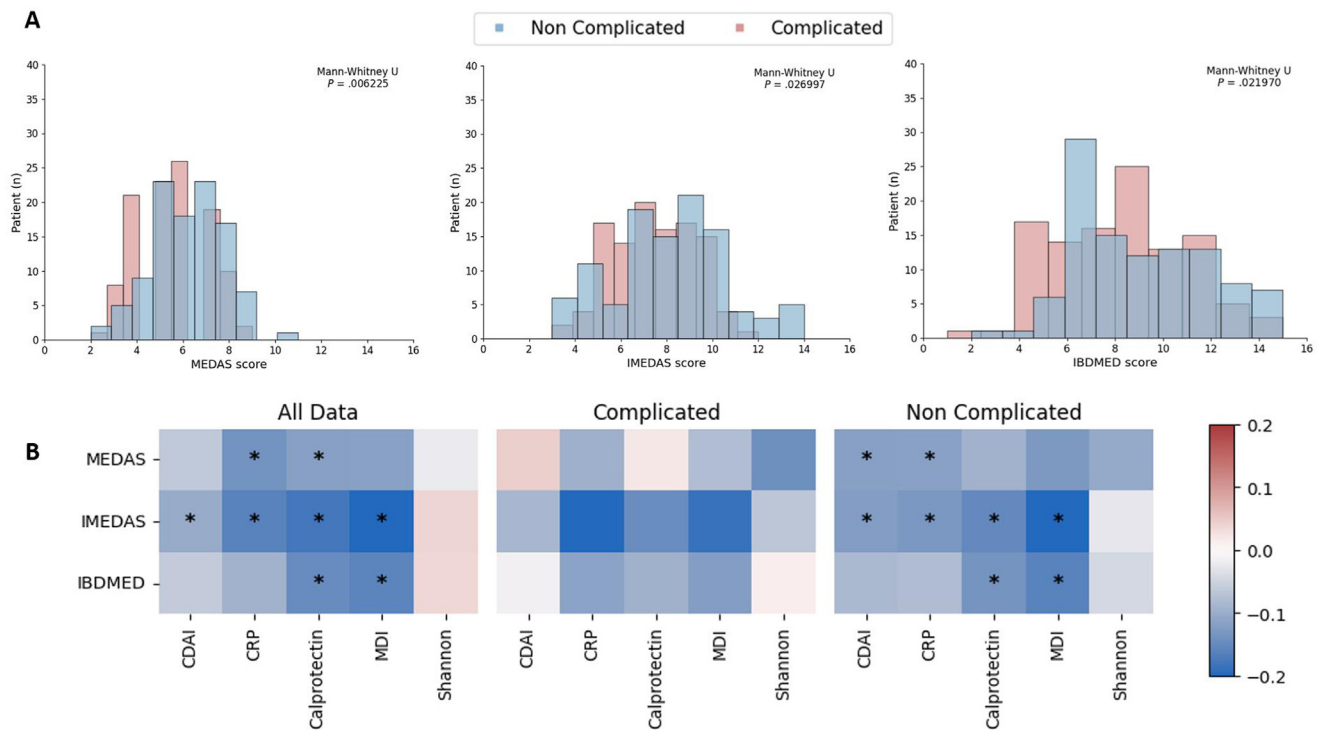
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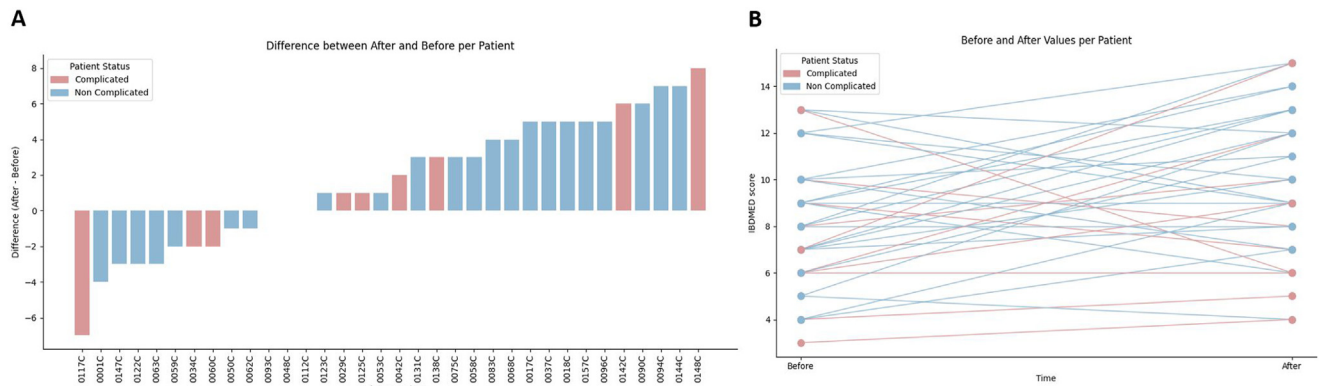
We thank the study participants for their effort and time.



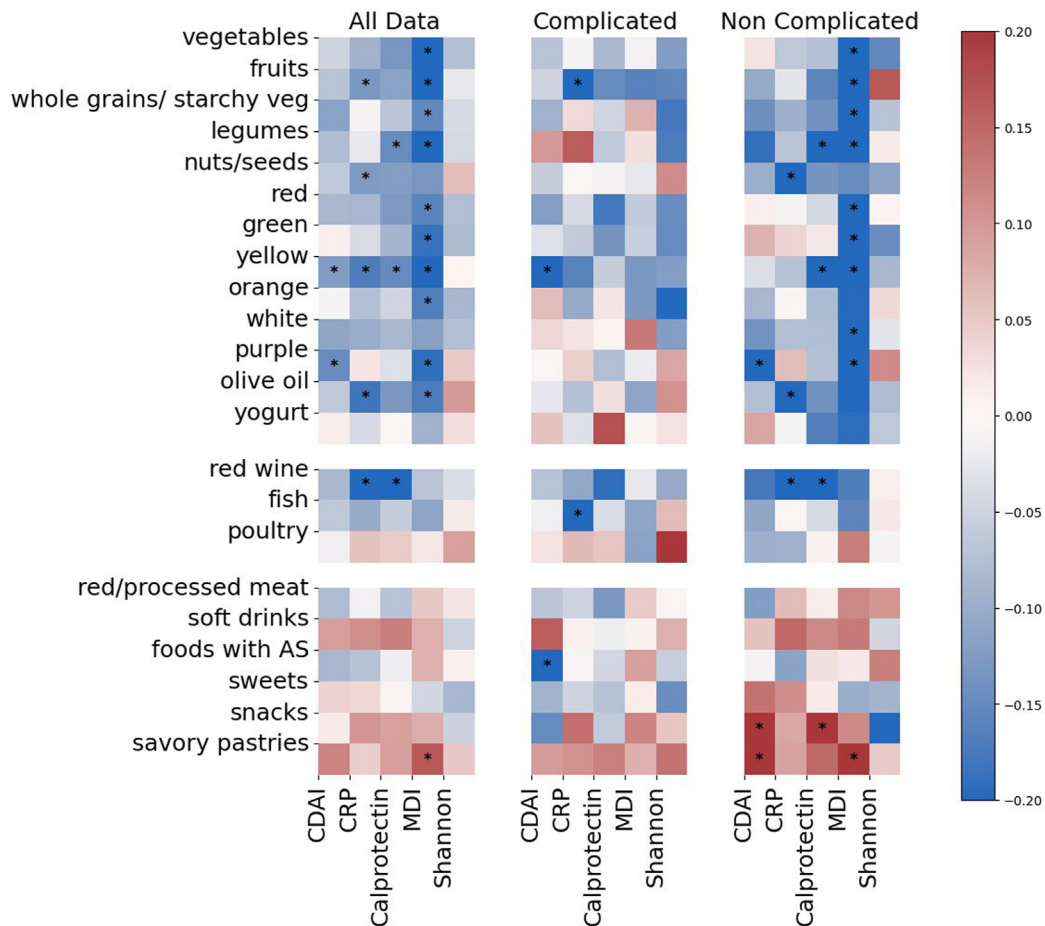
Supplementary Figure 1. Correlations between MED adherence scores (n = 271, FFQ = 636; $P < .05$ for all).



Supplementary Figure 2. Baseline adherence to MED is inversely associated with clinical, inflammatory, and microbial dysbiosis markers in patients with newly diagnosed CD. (A) Distribution of baseline MED adherence scores among patients with complicated vs noncomplicated CD course. (B) Spearman's correlations between MED adherence scores and clinical (CDAI), inflammatory (CRP, calprotectin), and microbial (MDI, Shannon diversity) scores across patients with complicated vs indolent CD course. * $P < .05$.



Supplementary Figure 3. Changes in the IBDMED score before and after dietary counseling. (A) Deltas in the IBDMED score before and after dietary counseling in a subset of patients with electronically documented dietary counseling and FFQs before and after dietary counseling ($n = 37$). For patients with more than 2 FFQs, we compared the maximal scores before and after the counseling. (B) Changes in IBDMED scores before and after dietary counseling. Paired t test, $P = .008$.



Supplementary Figure 4. Baseline consumption of MED-recommended dietary components is inversely associated with clinical, inflammatory, and dysbiosis markers in patients with newly diagnosed CD. Spearman's correlations between MED-recommended and non-recommended dietary components (derived only from baseline data) and clinical (CDAI), inflammatory (CRP, calprotectin), and microbial (MDI, Shannon diversity) scores across patients with complicated vs noncomplicated CD course. $*P < .05$. AS, artificial sweeteners.